

# Matching Division Stories, Drawings, and Equations

## Answer Key

Grades 2–3    Operations and Algebraic Thinking

### Quick Answers

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|      |      |      |        |
|------|------|------|--------|
| 1. 4 | 4. 4 | 7. 4 | 10. 9  |
| 2. 5 | 5. 3 | 8. 4 | 11. 5  |
| 3. 6 | 6. 6 | 9. 7 | 12. 10 |

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### Common wrong answers

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4. *If they got 18: subtracted the two numbers instead of dividing*  
8. *If they got 256: multiplied the two numbers instead of dividing*  
10. *If they got 4: counted the rows as the answer instead of dividing to find the number in each row*  
12. *If they got 2: divided one pack by the tables and forgot the other packs*
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#### 1. 12 stickers shared by 3 friends.

**Step 1.** Name the three parts before writing anything. The total in all is 12 and the number of groups is 3; what is unknown is how many are in each group. That is the sharing shape of division.

**Step 2.** Write it as total  $\div$  groups:  $12 \div 3$ , and count out 12 into 3 equal piles — 4 in each pile. 4

*Check with multiplication:*  $3 \times 4 = 12$ , which is the same fact read the other way.

#### 2. The picture of 20 counters in 4 equal groups.

**Step 1.** A drawing already shows two of the three parts: count the total (20 counters) and count the boxes (4 groups). The unknown is the size of one group.

**Step 2.**  $20 \div 4$ : each box holds the same number, and counting one box gives 5. 5

*Matching multiplication:*  $4 \times 5 = 20$  — 4 groups of 5.

#### 3. $3 \times \square = 18$ and its division partner.

**Step 1.** A missing *factor* is exactly what division finds, so turn the multiplication around instead of guessing.

**Step 2.** The partner sentence is  $18 \div 3 = \square$ . Skip-count by threes to 18: 3, 6, 9, 12, 15, 18 — that is 6 threes. 6

*Fact family:*  $3 \times 6 = 18$ ,  $6 \times 3 = 18$ ,  $18 \div 3 = 6$ ,  $18 \div 6 = 3$ .

#### 4. 24 crayons packed 6 to a box.

**Step 1.** This story gives the total (24) and the size of each group (6 in a box). The unknown is the *number* of groups, so it is the measurement shape of division, not the sharing shape.

**Step 2.**  $24 \div 6$ : take 6 away at a time — 24, 18, 12, 6, 0. That is 4 boxes. 4

*Watch out:* a student who reads “24 and 6” without reading the story may subtract and write 18. Subtraction takes 6 away *once*; division takes 6 away until nothing is left.

**5. Which story matches  $15 \div 5 = \square$ ?**

**Step 1.** Read the equation out loud first: “15 in all, split into 5 equal parts.” Now look for the story that starts with 15 as the total.

**Step 2.** Story **A** is the match — 15 shells split into 5 equal piles. Story B makes 15 the size of each group, so its total would be much larger than 15. Answering A:  $15 \div 5 = 3$  shells in each pile. 3

*Language clue:* “put into 5 equal piles” names groups; “15 in each” names group size.

**6. The picture of 18 counters in 3 equal groups.**

**Step 1.** Read the same two facts off the drawing: 18 counters in all, 3 boxes. The unknown is the size of one group.

**Step 2.**  $18 \div 3$ : each box holds the same amount, and one box holds 6. 6

*Matching multiplication:*  $3 \times 6 = 18$ . The same picture tells both sentences — multiplication puts the groups together, division breaks them apart.

**7. The fact family of 7, 4 and 28.**

**Step 1.** A missing factor again, so use the division partner rather than guessing and checking.

**Step 2.**  $28 \div 7$ : count by sevens — 7, 14, 21, 28. That is 4 sevens. 4

*All four sentences:*  $7 \times 4 = 28$ ,  $4 \times 7 = 28$ ,  $28 \div 7 = 4$ ,  $28 \div 4 = 7$ . Every one uses the same three numbers.

**8. 32 apples in baskets of 8.**

**Step 1.** Decide which number is the total before writing the equation. “A farm stand has 32 apples” is the whole amount; “8 apples in each basket” is the group size. The total is always the number you divide.

**Step 2.**  $32 \div 8$ : count by eights — 8, 16, 24, 32. That is 4 baskets. 4

*Watch out:* multiplying gives 256, which is more apples than the stand ever had. If an answer is bigger than the total, the operation was wrong.

**9. Which story matches  $42 \div 6 = \square$ ?**

**Step 1.** Say the equation in words: “42 in all, put into groups of 6.” The total must be 42.

**Step 2.** Story **A** matches — 42 muffins with 6 on each tray. Story B says 42 muffins are on *each* of 6 trays, so 42 is a group size there and the total would be far more than 42. Answering A:  $42 \div 6 = 7$  trays. 7

*Why B fails:* B describes multiplication,  $6 \times 42$ , not division.

**10. The array of 36 counters in 4 rows.**

**Step 1.** An array gives the total (36) and the number of rows (4). The question asks for the size of a row, so divide the total by the number of rows.

**Step 2.**  $36 \div 4$ : count by fours to 36 — 4, 8, 12, 16, 20, 24, 28, 32, 36. That is 9, so each row holds 9 counters. 9

*The trap:* answering 4 would be reporting the number of *rows*, which the picture already told you. Read the question, then decide which part of the array is unknown.

**11.  $45 \div \square = 9$  — the divisor is missing.**

**Step 1.** Here the total (45) and the answer (9) are known, so rewrite the sentence as multiplication:  $\square \times 9 = 45$ .

**Step 2.** Count by nines — 9, 18, 27, 36, 45 — which takes 5 nines. 5

*A story it could tell:* 45 marbles are shared equally among 5 children, and each child gets 9.

**12. Challenge: 5 packs of 6 pencils shared by 3 tables.**

**Step 1.** You cannot divide yet, because the total is not given — only the packs are. Find the total first, then share it.

**Step 2.** Total pencils:  $5 \times 6 = 30$ .

**Step 3.** Now share the total among the tables:  $30 \div 3 = 10$  pencils for each table. 10

*The trap:* dividing  $6 \div 3$  gives 2 and only shares *one* pack. A picture of 3 groups with 10 pencils in each shows the whole 30.